

PROTECTION PRODUCTS - RailClamp®

Description

RailClamp® TVS diodes are specifically designed to protect sensitive components which are connected to high-speed data and transmission lines from overvoltage caused by ESD (electrostatic discharge), CDE (cable discharge events), and EFT (electrical fast transients).

The RClamp®0502BA has a typical capacitance of only 0.60pF (pin 1 to 2). This means it can be used on circuits operating in excess of 3GHz with minimal signal attenuation. They may be used to meet the ESD immunity requirements of IEC 61000-4-2, Level 4 ($\pm 15\text{kV}$ air, $\pm 8\text{kV}$ contact discharge). Each device can be configured to protect 1 bidirectional line or two unidirectional lines.

These devices are in a small SC-75 (SOT-523) package and feature a lead-free, matte tin finish. They are compatible with both lead free and SnPb assembly techniques. They are designed for use in applications where board space is at a premium. The combination of small size, low capacitance, and high level of ESD protection makes them a flexible solution for applications such as HDMI, MDDI, antenna circuits, Automatic Test Equipment, USB 2.0, and Infiniband circuits.

Features

- ◆ Transient protection for high-speed data lines to **IEC 61000-4-2 (ESD) $\pm 18\text{kV}$ (air), $\pm 12\text{kV}$ (contact)**
IEC 61000-4-4 (EFT) 40A (5/50ns)
- ◆ Protects up to two I/O lines
- ◆ Ultra-Low capacitance (**$< 1\text{pF}$**)
- ◆ No insertion loss to **$> 3.0\text{GHz}$**
- ◆ Low profile ($< 1\text{mm}$)
- ◆ Low leakage current and clamping voltage
- ◆ Low operating voltage: 5.0V
- ◆ Solid-state silicon-avalanche technology

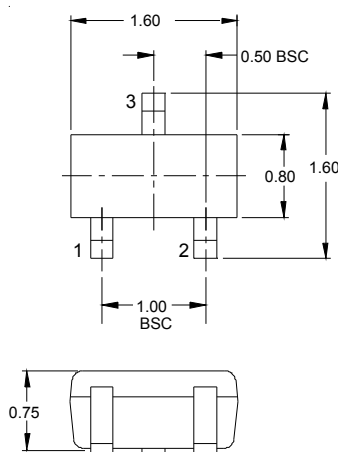
Mechanical Characteristics

- ◆ SC-75 (SOT-523) package
- ◆ Lead Finish: Matte Tin
- ◆ Pb-Free, Halogen Free, RoHS/WEEE Compliant
- ◆ Molding compound flammability rating: UL 94V-0
- ◆ Packaging: Tape and Reel

Applications

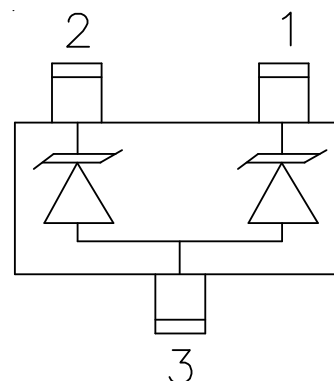
- ◆ Mobile Display Digital Interface (MDDI)
- ◆ USB 2.0
- ◆ Firewire Ports
- ◆ GaAs Photodetector Protection
- ◆ HBT Power Amp Protection
- ◆ Infiniband Transceiver Protection

Dimensions



Nominal Dimensions (mm)

Schematic & PIN Configuration



SC-75 3L (Top View)

PROTECTION PRODUCTS
Absolute Maximum Rating

Rating	Symbol	Value	Units
Peak Pulse Power ($t_p = 8/20\mu s$)	P_{pk}	125	Watts
Peak Pulse Current ($t_p = 8/20\mu s$)	I_{pp}	5	A
ESD per IEC 61000-4-2 (Air) ESD per IEC 61000-4-2 (Contact)	V_{ESD}	18 12	kV
Operating Temperature	T_J	-55 to +125	°C
Storage Temperature	T_{STG}	-55 to +150	°C

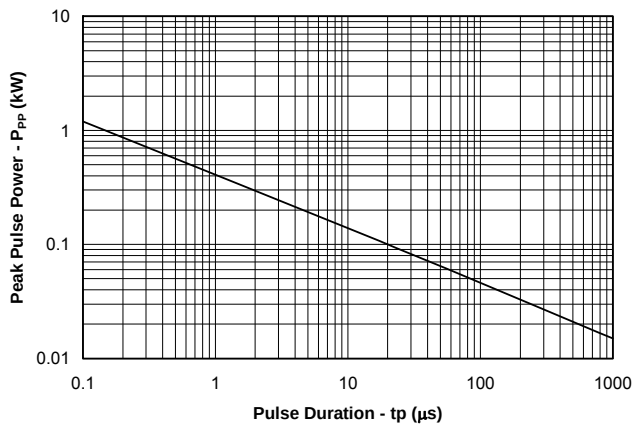
Electrical Characteristics (T=25°C)

Parameter	Symbol	Conditions	Minimum	Typical	Maximum	Units
Reverse Stand-Off Voltage	V_{RWM}	Pin 1 or Pin 2 to Pin 3 and Between Pins 1 and 2			5	V
Reverse Breakdown Voltage	V_{BR}	$I_t = 1mA$ Pin 1 or Pin 2 to Pin 3 and Between Pins 1 and 2	6			V
Reverse Leakage Current	I_R	$V_{RWM} = 5V, T=25^\circ C$ Pin 1 or Pin 2 to Pin 3 and Between Pins 1 and 2			1	μA
Clamping Voltage	V_C	$I_{pp} = 1A, t_p = 8/20\mu s$ Pin 1 to Pin 2			15	V
Clamping Voltage	V_C	$I_{pp} = 5A, t_p = 8/20\mu s$ Pin 1 or Pin 2 to Pin 3			22	V
Clamping Voltage	V_C	$I_{pp} = 5A, t_p = 8/20\mu s$ Pin 1 to Pin 2			25	V
Junction Capacitance	C_j	$V_R = 0V, f = 1MHz$ Pin 1 to Pin 2		0.60	0.9	pF
Junction Capacitance	C_j	$V_R = 0V, f = 1MHz$ Pin 1 or Pin 2 to Pin 3			1.2	pF

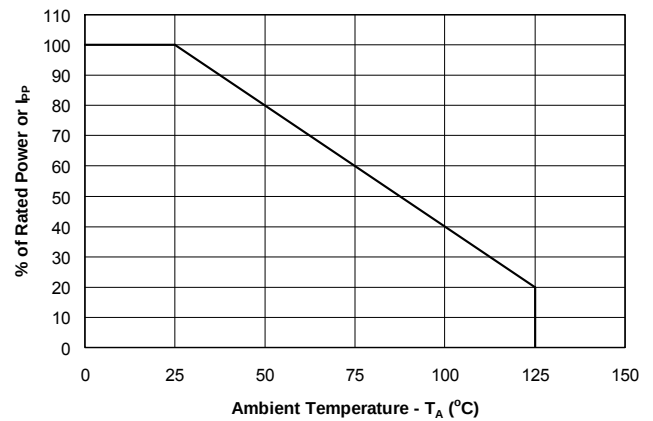
PROTECTION PRODUCTS

Typical Characteristics

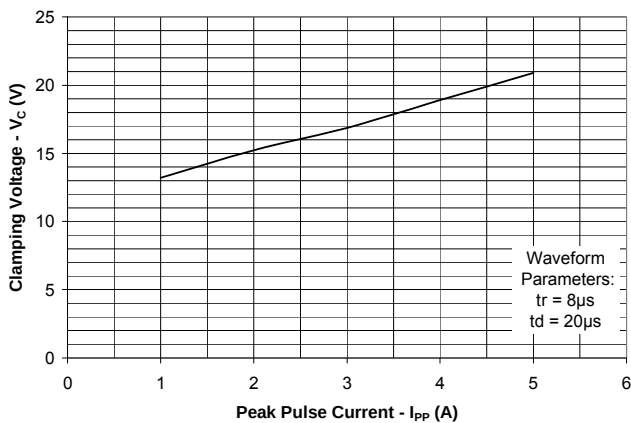
Non-Repetitive Peak Pulse Power vs. Pulse Time



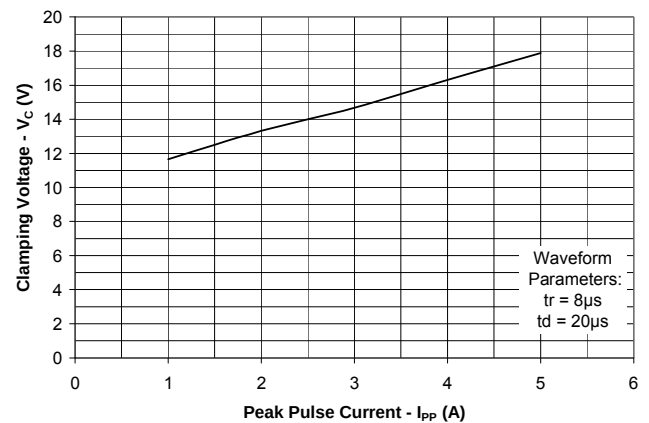
Power Derating Curve



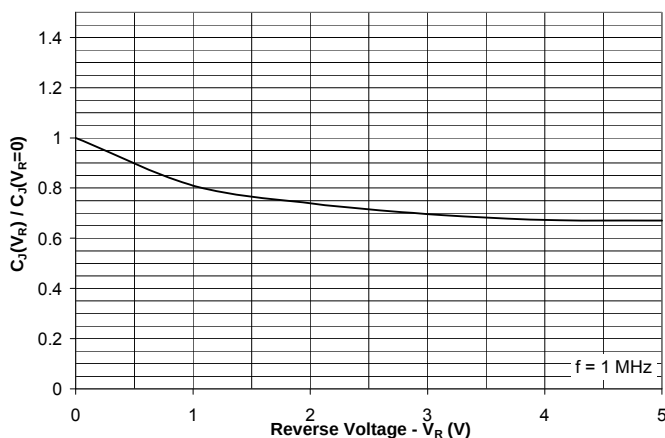
**Clamping Voltage vs. Peak Pulse Current
Pin 1 to Pin 2**



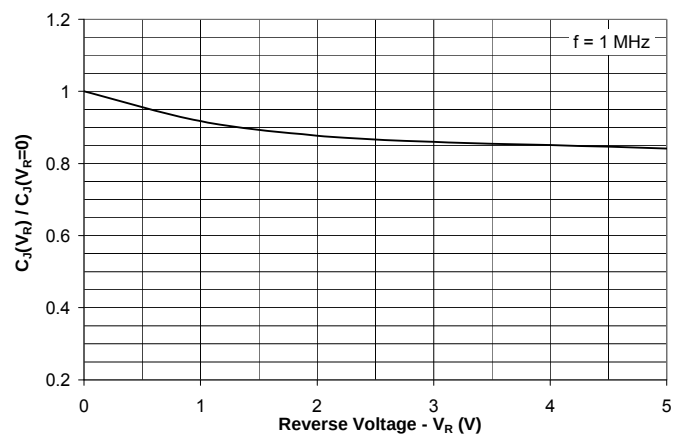
**Clamping Voltage vs. Peak Pulse Current
Pin 1 or Pin 2 to Pin 3**



**Normalized Capacitance vs. Reverse Voltage
Pin 1 or Pin 2 to Pin 3**



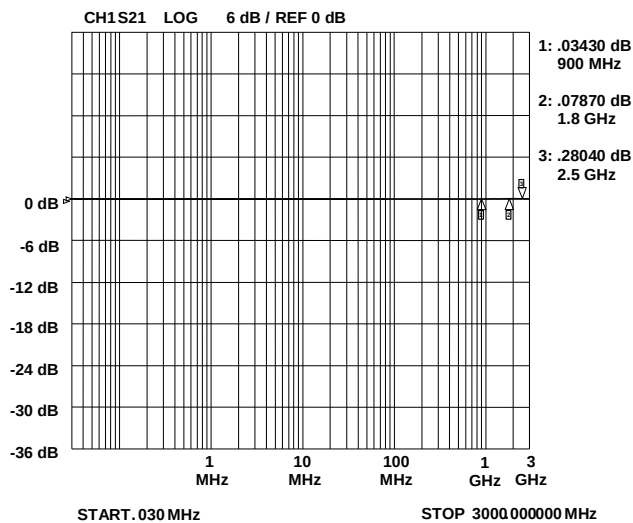
**Normalized Capacitance vs. Reverse Voltage
Pin 1 to Pin 2**



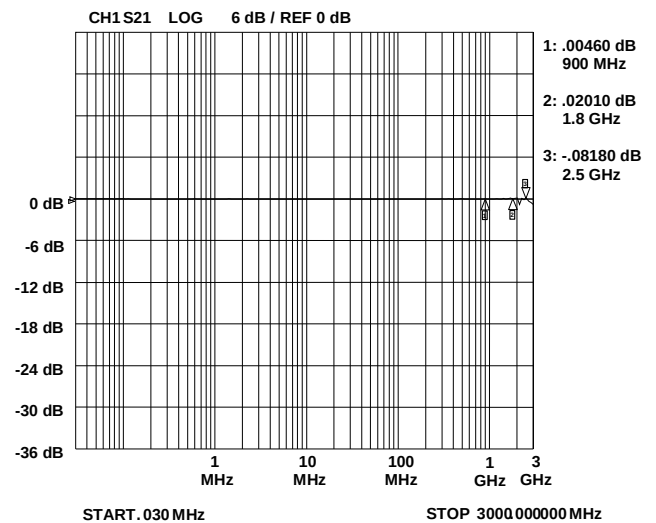
PROTECTION PRODUCTS

Typical Characteristics

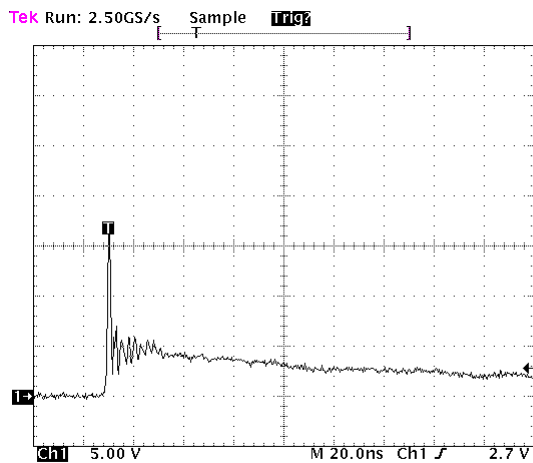
Insertion Loss S21 (Pin 1 to Pin 2)



Insertion Loss S21 (Pin 1 or Pin 2 to Pin 3)

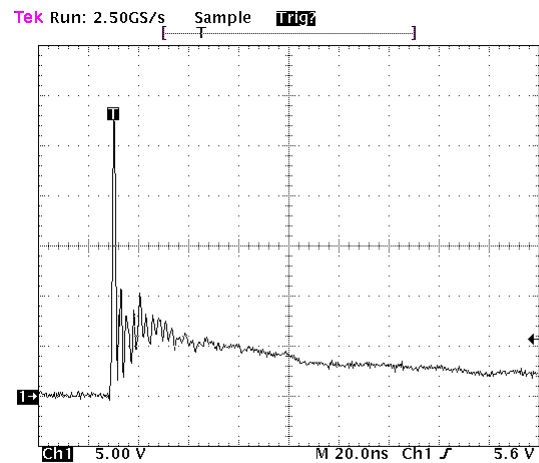


ESD Clamping (4kV Contact per IEC 61000-4-2)



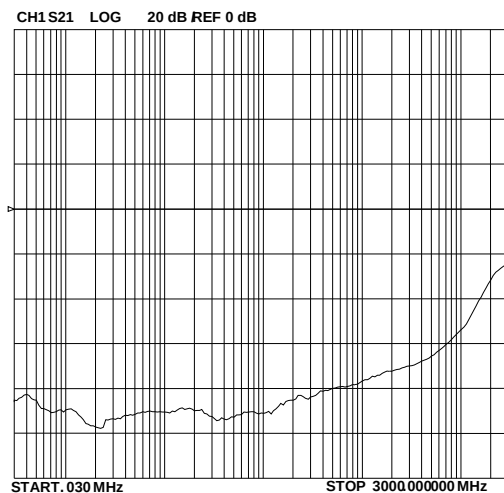
Note: Data is taken with a 10x attenuator

ESD Clamping (8kV Contact per IEC 61000-4-2)



Note: Data is taken with a 10x attenuator

Analog Crosstalk



PROTECTION PRODUCTS**Applications Information****Device Connection Options**

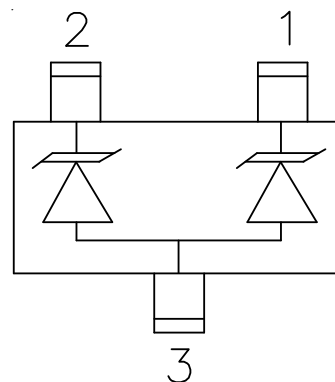
This device is optimized for protection of 1 line operating in excess of 3GHz. It may also be used to protect two lines operating in excess of 2.0GHz. The device is connected as follows:

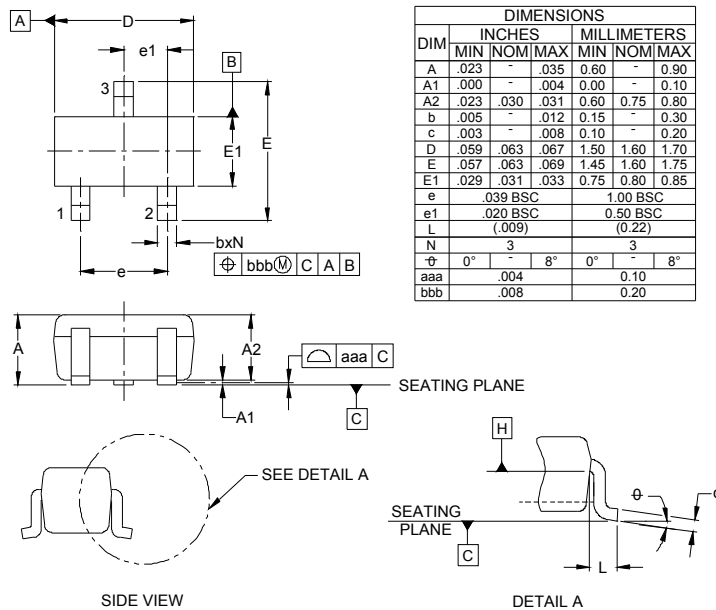
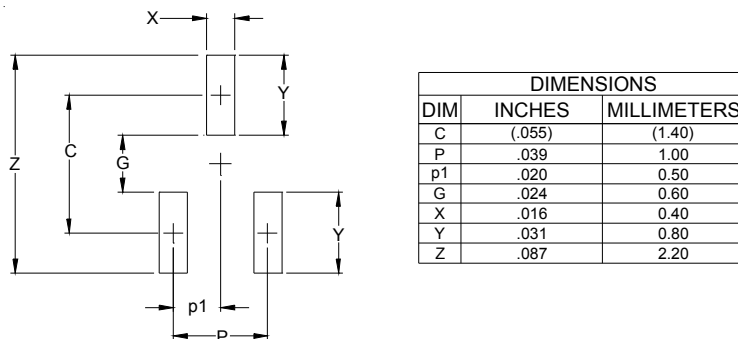
Protection for one line with $<1\text{pF}$ capacitance can be achieved by connecting one data line to either pin 1 or pin 2 with the other pin connected to ground. Pin 3 is not connected. The connection to ground should be made directly to a ground plane. The path length should also be kept as short as possible to minimize parasitic inductance.

Protection of two lines is achieved by connecting data lines at pins 1 & 2. Pin 3 is connected to ground. The connection to ground should be made directly to a ground plane. The path length should also be kept as short as possible to minimize parasitic inductance.

Matte Tin Lead Finish

Matte tin has become the industry standard lead-free replacement for SnPb lead finishes. A matte tin finish is composed of 100% tin solder with large grains. Since the solder volume on the leads is small compared to the solder paste volume that is placed on the land pattern of the PCB, the reflow profile will be determined by the requirements of the solder paste. Therefore, these devices are compatible with both lead-free and SnPb assembly techniques. In addition, unlike other lead-free compositions, matte tin does not have any added alloys that can cause degradation of the solder joint.

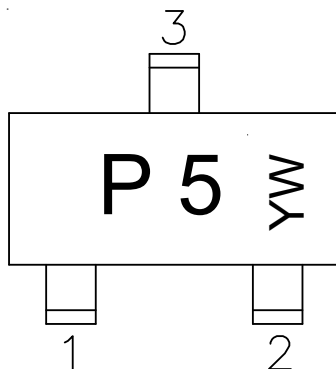
Figure 1. Pin Configuration

PROTECTION PRODUCTS
Outline Drawing -SC-75 (SOT-523)

Land Pattern -SC-75 (SOT-523)

NOTES:

- THIS LAND PATTERN IS FOR REFERENCE PURPOSES ONLY
CONSULT YOUR MANUFACTURING GROUP TO ENSURE YOUR
COMPANY'S MANUFACTURING GUIDELINES ARE MET.

PROTECTION PRODUCTS

Marking



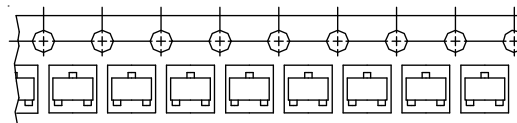
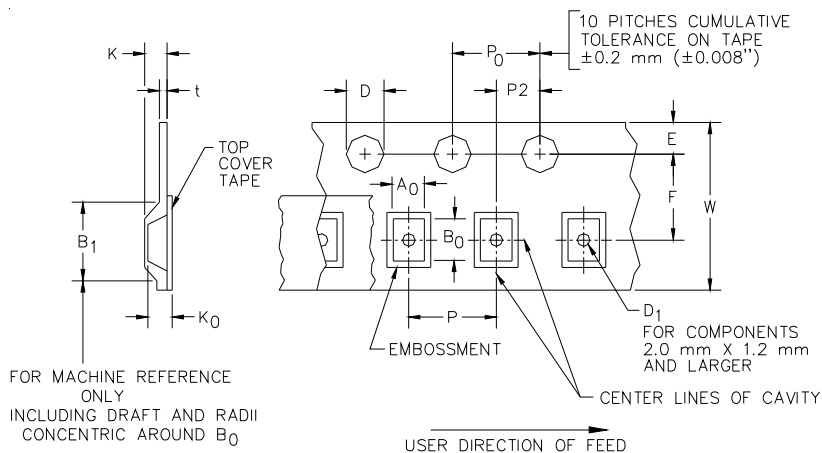
YW = Date Code

Ordering Information

Part Number	Lead Finish	Qty per Reel	Reel Size
RClamp0502BATCT	Pb Free	3,000	7 Inch

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Tape and Reel Specification



Device Orientation in Tape

A0	B0	K0
1.85 +/-0.15 mm	1.85 +/-0.15 mm	0.87 +/-0.15 mm

Tape Width	B, (Max)	D	D1 (MIN)	E	F	K (MAX)	P	P0	P2	T(MAX)	W
8 mm	4.2 mm (.165)	1.5 + 0.1 mm - 0.0 mm (0.59 +.005 - .000)	1.0 mm (.039)	1.750±.10 mm (.069±.004)	3.5±0.05 mm (.138±.002)	2.4 mm (.094)	4.0±0.1 mm (.157±.004)	4.0±0.1 mm (.157±.004)	2.0±0.05mm (.079±.002)	0.4 mm (.016)	8.3 mm (.312±.012)

Contact Information

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